

Exploring Real Data with CODAP

CODAP · Concord Consortium

Senior Secondary (Years 11-12)

~30 minutes

Mathematics · General
Mathematics

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://codap.concord.org/app/?v=3>

Curriculum links: bivariate data · data analysis · statistical inference · statistics

Learning intentions

- I can use CODAP to construct appropriate graphical displays for univariate and bivariate data.
- I can describe the shape, centre and spread of a distribution from a graph I have built.
- I can draw a supported inference from a real dataset.

Getting started

Open CODAP and either import a dataset of your own (as a CSV) or use one of the built-in example datasets from the 'File' or 'Example Documents' menu. Drag an attribute (column) from the case table onto the x-axis of a new graph to begin, then drag a second attribute onto the y-axis to explore relationships between two variables.

Tasks

1. Load a dataset into CODAP. List the attributes (variables) available and identify which ones are numerical and which are categorical.

2. Drag one numerical attribute onto a graph's x-axis to produce a dot plot or histogram. Describe the shape of the distribution (symmetric, skewed, or uniform) and note any outliers.

3. Use CODAP's summary tools (or your own calculation) to find the mean, median and standard deviation or range of your chosen numerical attribute.

Working:

4. Choose two numerical attributes and drag one onto the x-axis and the other onto the y-axis to create a scatterplot. Describe the strength and direction of any relationship you see.

5. Test three different pairs of attributes as scatterplots. Record what kind of relationship each pair shows.

Attribute pair	Relationship direction	Strength

6. Use CODAP's 'least squares line' or trend line tool (if available) on your strongest scatterplot from Task 5. Interpret the slope of the line in the context of your dataset.

7. Split your data into two groups using a categorical attribute (drag it onto the graph to colour or separate the points). Compare the distributions of a numerical attribute between the two groups and describe any differences.

8. Based on your analysis in this worksheet, write a short statistical conclusion about your dataset, including at least one claim supported by evidence from your graphs, and one limitation of your data or method.

Extension (optional)

Data journalists often present bivariate relationships as 'X causes Y' based on a strong correlation. Using one of your scatterplots, explain why a strong correlation between two variables in your dataset does not necessarily prove that one causes the other.
